# Human Neural Network guide

5 minutes · squads of 5 · standing

## The idea

A real AI model guesses from clues. Your squad becomes a tiny model: two scouts see a card and call out clues (the Input layer), two weigh the clues and vote (the Hidden layer), and one announces the guess (the Output layer). The model only knows what it was taught, so a misleading card shows how AI gets fooled.

## Print and prep, per squad

- Page 2: cut out the 5 role badges and the clue word bank. Page 3: cut the 12 mystery cards (cards 1 and 2 are the training examples). Page 4: one scorecard.
- Shuffle nothing. The cards are already in play order, and card 12 (the red ball) is the trick card. Keep the answer key below to yourself.
- Two classes the model must learn: A = FRUIT and B = BALL.

## Roles (squad stands in a line)

- INPUT 1 and INPUT 2 see the card and each call out ONE clue word, quietly, from the word bank.
- HIDDEN 1 and HIDDEN 2 do not see the card. They weigh the two clues and whisper a vote, A or B, to Output.
- OUTPUT hears both votes and announces the squad's guess: Fruit or Ball.

## How to run it

- Train the model first (30 seconds): show the whole squad cards 1 and 2 face up. Card 1 is Fruit (A) and card 2 is Ball (B). These two are ALL the model has ever seen. Put them away.
- Rounds 1 to 9 (cards 3 to 11): you flip a card to the Input scouts only. They give clues, Hidden votes, Output guesses. You say right or wrong. Record it on the scorecard. Rotate roles every 3 rounds. Aim for about 20 seconds a round.
- Round 10 (card 12), the trick card: the red ball. Round, red, shiny, so the model will probably guess Fruit and be wrong. Do not warn them.
- Ask: why did the model get it wrong? What clues were missing, like bounces, no stem, no leaf?

## Debrief (the last minute)

- A model only knows what its data showed it. If every training fruit was red and round, red and round means fruit.
- Connect to Level 7: when you build your Teachable Machine, vary your samples so it cannot rely on one misleading clue.
- Connect to Level 4: if the training data is unfair or too narrow, the model's guesses are too.

## Answer key (card number: answer)

1: Apple (Fruit) 2: Soccer ball (Ball) 3: Banana (Fruit) 4: Basketball (Ball) 5: Lemon (Fruit) 6: Tennis ball (Ball) 7: Orange (Fruit) 8: Beach ball (Ball) 9: Cherries (Fruit) 10: Baseball (Ball) 11: Strawberry (Fruit) 12: Red ball (Ball)

# Role badges and clue word bank

cut out · one set per squad

## INPUT 1

You see the card.
Call out ONE clue word.

## INPUT 2

You see the card.
Call out ONE clue word.

## HIDDEN 1

You do NOT see the card.
Hear both clues. Vote A or B.

## HIDDEN 2

You do NOT see the card.
Hear both clues. Vote A or B.

## OUTPUT

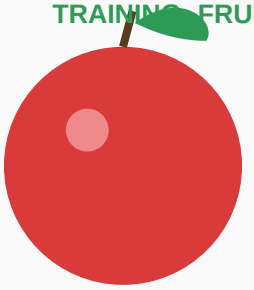
Hear both votes.
Announce the guess:
FRUIT or BALL.

### CLUE WORD BANK (Input scouts, pick ONE word per card)

| | | | | | |
|---|---|---|---|---|---|
| **round** | **red** | **shiny** | **has a stem** | **has a leaf** | **curved** |
| **stripes** | **seams** | **bounces** | **yellow** | **orange** | **patches** |
| **sweet** | **hard** | **smooth** | **bumpy** | **white** | **green** |

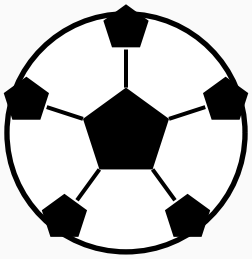
**MYSTERY CARDS: 12 per squad. Only the Input scouts see the front. Cut on the dotted lines.**

#1 TRAINING: FRUIT (A)



A.I.R.A. mystery card

#2 TRAINING: BALL (B)



A.I.R.A. mystery card

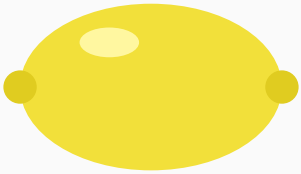
#3

A.I.R.A. mystery card

#4

A.I.R.A. mystery card

#5



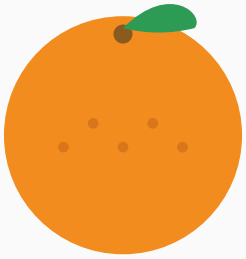
A.I.R.A. mystery card

#6



A.I.R.A. mystery card

#7

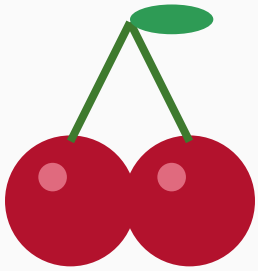


A.I.R.A. mystery card

#8

A.I.R.A. mystery card

#9

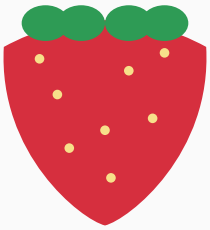


A.I.R.A. mystery card

#10

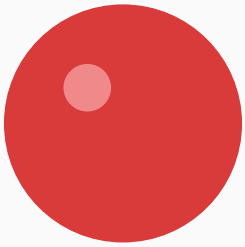
A.I.R.A. mystery card

#11



A.I.R.A. mystery card

#12



A.I.R.A. mystery card

# Squad scorecard

one per squad

**Squad:** _____

| Card # | Clue 1 | Clue 2 | Hidden 1 | Hidden 2 | Output guess | Right? |
|---|---|---|---|---|---|---|
| 3 | | | | | | |
| 4 | | | | | | |
| 5 | | | | | | |
| 6 | | | | | | |
| 7 | | | | | | |
| 8 | | | | | | |
| 9 | | | | | | |
| 10 | | | | | | |
| 11 | | | | | | |
| 12 | | | | | | |

## Reflect (card 12, the red ball)

What did the model guess, and was it right?

Which clue fooled the model? What clue was missing?

What would you add to the training data so it stops making this mistake?